

5 - Advanced Clustering Techniques

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Advanced Topics in Artificial Intelligence

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Gaussian Mixture Models (GMM)

Subsection 1

Definitions

Overview of GMM

- Gaussian Mixture Models (GMMs) are a type of unsupervised learning model that assumes all the data points are generated from a mixture of several Gaussian distributions with unknown parameters, each representing one cluster or component. These Gaussian distributions are blended together to form the complete dataset.
- GMMs are used for clustering similar to k-means but provide more flexibility due to the incorporation of distributional information. In particular, GMM provides a probabilistic clustering approach, allowing soft assignments.
- GMMs can accommodate clusters that have different sizes and correlations as well as different shapes, unlike k-means which assumes all clusters are spherical.
- Each Gaussian component is characterized by mean (μ), covariance matrix (Σ), and mixing coefficient (π).

Overview of GMM

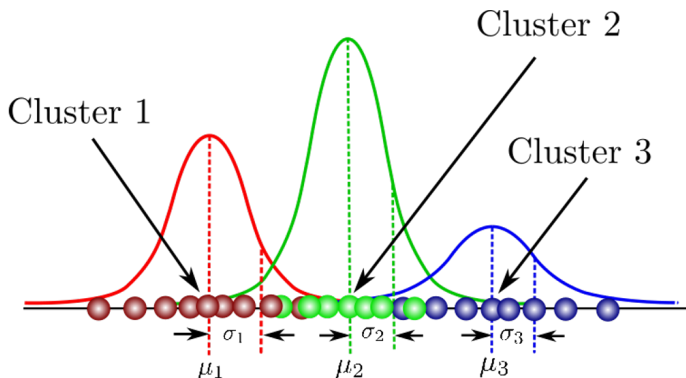


Figure 1: A simple example with three Gaussian functions, each Gaussian explains the data contained in each of the three clusters.

Overview of GMM

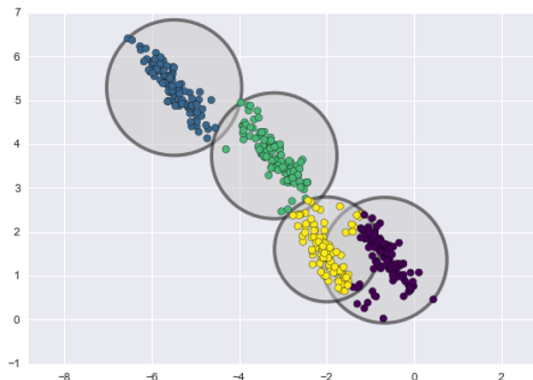


Figure 2: All the instances generated from a single Gaussian distribution form a cluster that typically looks like an ellipsoid. Each cluster can have a different ellipsoidal shape, size, density and orientation

Mathematical Formulation

- The probability density function (PDF) of a Gaussian mixture model is given by:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k) \quad (1)$$

- Where:

- π_k are the mixing coefficients such that $\sum \pi_k = 1$.
- $\mathcal{N}(x | \mu_k, \Sigma_k)$ is the Gaussian distribution defined as:

$$\mathcal{N}(x | \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right) \quad (2)$$

where d is the dimensionality of x .

Subsection 2

Expectation-Maximization Algorithm

Expectation-Maximization (EM) Algorithm

To estimate the parameters (means, covariances, and mixture weights) of the Gaussian components, GMMs typically use the **Expectation-Maximization (EM)** algorithm. The EM algorithm iteratively performs two steps until convergence:

- **Expectation Step (E-step):** Given the current parameters, compute the probability that each data point belongs to each Gaussian component. This step essentially assigns weights to each data point based on how likely they are to belong to each component.
- **Maximization Step (M-step):** Update the parameters of the Gaussians to maximize the likelihood of the data given these new assignments.

E-step: Computing Responsibilities

- The first step, known as the **Expectation step** or **E step**, consists of calculating the expectation of the component assignments C_k for each data point $x_i \in X$ given the model parameters π_k , μ_k and σ_k (the probability of each data point belonging to each distribution).
- For each point x_i compute r_{ic} the probability that x_i belong to the cluster c

$$r_{ic} = \frac{\pi_c \mathcal{N}(x_i | \mu_c, \Sigma_c)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)} \quad (3)$$

E-step: Computing Responsibilities

Initialization of Parameters

- In the Expectation-Maximization (EM) algorithm, the Expectation step (E-step) requires the posterior probabilities of each cluster given the data.
- However, in the **first iteration**, we don't yet have estimates for the parameters (μ_k, Σ_k, π_k) .
- So, how do we start?

E-step: Computing Responsibilities

Initialization of Parameters

Since we don't know anything about the Gaussian distributions at the start, we need to **initialize** the parameters:

- **Means (μ_k)**: Usually initialized randomly or using K-Means clustering to get rough centroids.
- **Covariance matrices (Σ_k)**: Often set to identity matrices or computed from initial clusters.
- **Mixing coefficients (π_k)**: Set to **equal probabilities**, meaning each Gaussian starts with weight $\pi_k = \frac{1}{K}$.

After initialization, we proceed to the **first E-step**.

M-step: Parameter Update

Once we compute the responsibilities in the first **E-step**, we update the parameters in the **M-step**:

- Updating the means:

$$\mu_k = \frac{\sum_{i=1}^N r_{ik} x_i}{\sum_{i=1}^N r_{ik}} \quad (4)$$

- Updating the covariance matrices:

$$\Sigma_k = \frac{\sum_{i=1}^N r_{ik} (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^N r_{ik}} \quad (5)$$

- Updating the mixing coefficients:

$$\pi_k = \frac{\sum_{i=1}^N r_{ik}}{N} \quad (6)$$

The parameters μ_k , Σ_k , and π_k are iteratively updated until convergence.

EM Algorithm

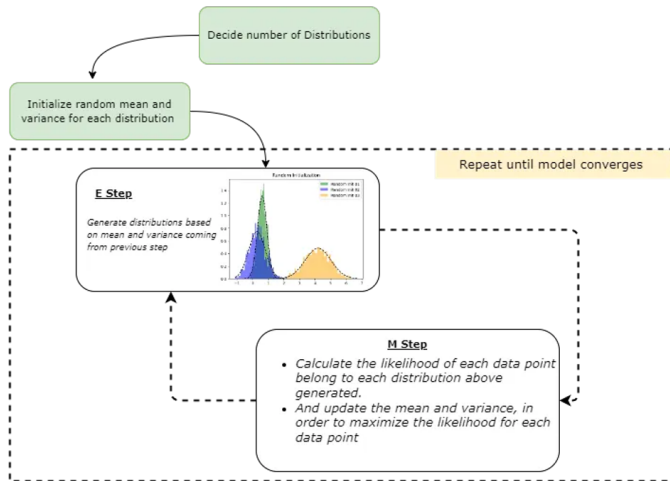


Figure 3: The Expectation-Maximization Process

Convergence of EM Algorithm

- The log-likelihood function is guaranteed to increase with each iteration.
- The algorithm stops when the change in log-likelihood is below a certain threshold:

$$|\log L^{(t)} - \log L^{(t-1)}| < \epsilon \quad (7)$$

- Convergence is usually fast, but EM can get stuck in local optima.
- Random initialization or multiple runs can help improve results.

Subsection 3

Advantages and Challenges of GMM

Advantages of GMM

- **Soft Clustering:** Unlike k-means, which assigns each data point to exactly one cluster, GMM provides probabilistic membership, allowing a more flexible clustering structure.
- **Modeling Complex Distributions:** GMM can model elliptical and overlapping clusters, which makes it superior to methods that assume spherical clusters, like k-means.
- **Handles Different Cluster Sizes and Shapes:** The covariance matrices allow clusters to have different orientations, densities, and scales.
- **Probabilistic Interpretation:** The output provides probabilities for each data point belonging to each cluster, which is useful in scenarios like anomaly detection and decision-making.
- **Adaptability to High-Dimensional Data:** GMM can work efficiently in high-dimensional feature spaces where simpler clustering algorithms may struggle.

Challenges of GMM

- **Sensitive to Initialization:** The algorithm's performance highly depends on initial parameter values. Poor initialization can lead to convergence to local optima.
- **Computational Complexity:** The Expectation-Maximization (EM) algorithm involves expensive matrix computations, particularly in high-dimensional spaces.
- **Prone to Overfitting:** If too many Gaussian components are used, GMM may fit noise in the data rather than underlying patterns.
- **Choosing the Number of Components:** Selecting the optimal number of Gaussians (K) is non-trivial and often requires model selection techniques like Bayesian Information Criterion (BIC) or Akaike Information Criterion (AIC).
- **Assumption of Gaussian Distributions:** GMM assumes that data is generated from a mixture of Gaussian distributions, which may not always be valid.

Mitigating Challenges

- **Improving Initialization:** Use k-means clustering or the K-means++ algorithm for better initialization of means before running EM.
- **Regularization:** Adding a small regularization term to covariance matrices can prevent singularities and overfitting.
- **Model Selection:** Use BIC and AIC to determine the optimal number of components, avoiding excessive model complexity.
- **Multiple Runs:** Running EM multiple times with different initializations can help avoid poor local optima.
- **Dimensionality Reduction:** Applying PCA or other dimensionality reduction techniques can help alleviate computational burdens in high-dimensional data.

Subsection 4

Selection and Evaluation of GMM

Choosing the Number of Components

- The number of Gaussian components (K) in GMM significantly affects performance.
- Common methods to determine K :
 - **Bayesian Information Criterion (BIC)**: Penalizes model complexity to avoid overfitting.
 - **Akaike Information Criterion (AIC)**: Measures model fit but does not penalize complexity as strongly as BIC.
 - **Cross-validation**: Splitting data into training and validation sets to evaluate model generalization.
 - **Elbow Method**: Observing the log-likelihood curve and selecting the point where improvement slows.

Choosing the Number of Components

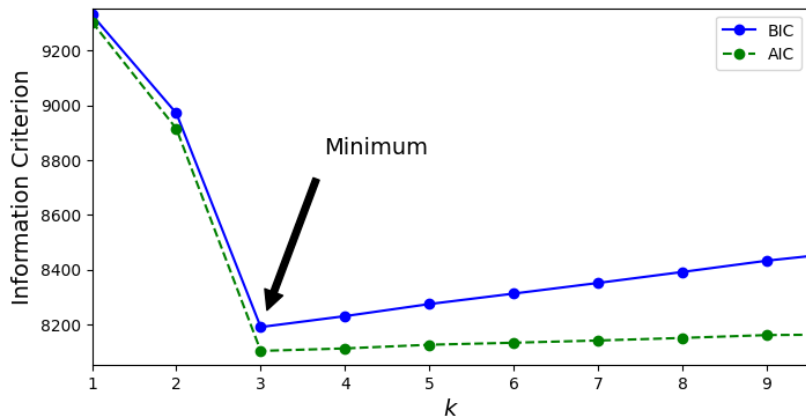


Figure 4: The application of the **Information Criterion** to find the optimal number of components

Evaluating GMM Performance

- **Log-Likelihood:** Measures how well the model explains the observed data.

$$L(\Theta) = \sum_{i=1}^N \log \sum_{k=1}^K \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k) \quad (8)$$

- **BIC and AIC Comparison:**

$$\text{BIC} = -2 \log L + K \log N, \quad \text{AIC} = -2 \log L + 2K \quad (9)$$

- **Cluster Quality Metrics:**

- **Silhouette Score:** Measures how similar a point is to its own cluster vs. other clusters.
- **Rand Index & Adjusted Rand Index:** Compare clustering results with ground truth labels.
- **Normalized Mutual Information (NMI):** Measures mutual dependence between clusters and true labels.

Visualization Techniques

- **Contour Plots:** Visualizing Gaussian distributions in 2D data.
- **Density Estimation:** Plotting probability density functions for each cluster.
- **Cluster Assignments:** Color-coding data points by their most probable cluster assignment.
- **Principal Component Analysis (PCA):** Reducing high-dimensional data to visualize clusters effectively.

Subsection 5

Conclusion

Summary

- GMM is a powerful clustering method using probabilistic modeling.
- EM algorithm is used for parameter estimation.
- Model selection is crucial to avoid overfitting.
- Widely used in various machine learning applications.